Annotated Bibliography

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Contents

Community Search

Bagrow: Evaluating local community methods in networks **bagrow2008evaluating**

James P Bagrow. "Evaluating local community methods in networks". In: *Journal of Statistical Mechanics: Theory and Experiment* 2008.05 (May 2008), P05001. ISSN: 1742-5468. DOI: 10.1088/1742-5468/2008/05/p05001.

Annotations: **05/21/26**: I don't like this paper (old, modularity focused). Benchmark graphs: generate a BA($N = 512$, $m_0 = 8$) graph, partition nodes into ground truth communities, and rewire inter-community edges to form $Q \approx 1/2$. Accuracy via NMI. Their alleged take-home message is that stopping rule is more important than agglomeration scheme, and should be the focus of future work.

Abstract: We present a new benchmarking procedure that is unambiguous and specific to local community finding methods, allowing one to compare the accuracy of various methods. We apply this to new and existing algorithms. A simple class of synthetic benchmark networks is also developed, capable of testing properties specific to these local methods.

Barbieri et al.: Efficient and effective community search **barbieri2015efficient**

Nicola Barbieri et al. "Efficient and effective community search". In: *Data Mining and Knowledge Discovery* 29.5 (June 2015), pp. 1406–1433. ISSN: 1573-756X. DOI: 10.1007/s10618-015-0422-1.

Annotations: **05/21/26**: k-core community search. Efficient means runtime; local search is desired. Effective means that we want to find small clusters. Seems to supercede Cui et al. [4]. Addressing efficiency, they propose ShellStruct (which finds maximal k-core). Addressing effectiveness, they propose a greedy heuristic that prunes k-core via a local search. They evaluate both single-node and multi-node queries.

Abstract: Community search is the problem of finding a good community for a given set of query vertices. One of the most studied formulations of community search asks for a connected subgraph that contains all query vertices and maximizes the minimum degree. All existing approaches to min-degree-based community search suffer from limitations concerning efficiency, as they need to visit (large part of) the whole input graph, as well as accuracy, as they output communities quite large and not really cohesive. Moreover, some existing methods lack generality: they handle only single-vertex queries, find communities that are not optimal in terms of minimum degree, and/or require input parameters. In this work we advance the state of the art on community search by proposing a novel method that overcomes all these limitations: it is in general more efficient and effective—one/two orders of magnitude on average, it can handle multiple query vertices, it yields optimal communities, and it is parameter-free. These properties are confirmed by an extensive experimental analysis performed on various real-world graphs.

Chu et al.: Finding the Best k in Core Decomposition: A Time and Space Optimal Solution **chu2020finding**

Deming Chu et al. "Finding the Best k in Core Decomposition: A Time and Space Optimal Solution". In: *2020 IEEE 36th International Conference on Data Engineering (ICDE)*. IEEE, Apr. 2020, pp. 685–696. DOI: 10.1109/icde48307.2020.00065.

Abstract: The mode of k-core and its hierarchical decomposition have been applied in many areas, such as sociology, the world wide web, and biology. Algorithms on related studies often need an input value of parameter k, while there is no existing solution other than manual selection. In this paper, given a graph and a scoring metric, we aim to efficiently find the best value of k such that the score of the k-core (or k-core set) is the highest. The problem is challenging because there are various community scoring metrics and the computation is costly on large datasets. With the well-designed vertex ordering techniques, we propose time and space optimal algorithms to compute the best k, which are applicable to most community metrics. The proposed algorithms can compute the score of every k-core (set) and can benefit the solutions

to other k-core related problems. Extensive experiments are conducted on 10 real-world networks with size up to billion-scale, which validates both the efficiency of our algorithms and the effectiveness of the resulting k-cores.

Cui et al.: Local search of communities in large graphs — cui2014local

Wanyun Cui et al. "Local search of communities in large graphs". In: *Proceedings of the 2014 ACM SIGMOD International Conference on Management of Data*. SIGMOD/PODS'14. ACM, June 2014, pp. 991–1002. DOI: 10.1145/2588555.2612179.

Annotations: **05/21/26**: k-core community search, with minimum size, with single node queries, k as an argument. Finding small communities is good for runtime.

Abstract: Community search is important in social network analysis. For a given vertex in a graph, the goal is to find the best community the vertex belongs to. Intuitively, the best community for a given vertex should be in the vicinity of the vertex. However, existing solutions use global search to find the best community. These algorithms, although straight-forward, are very costly, as all vertices in the graph may need to be visited. In this paper, we propose a local search strategy, which searches in the neighborhood of a vertex to find the best community for the vertex. We show that, because the minimum degree measure used to evaluate the goodness of a community is not monotonic, designing effciient local search solutions is a very challenging task. We present theories and algorithms of local search to address this challenge. The efficiency of our local search strategy is verifeid by extensive experiments on both synthetic networks and a variety of real networks with millions of nodes.

Fang et al.: A survey of community search over big graphs — fang2019survey

Yixiang Fang et al. "A survey of community search over big graphs". In: *The VLDB Journal* 29.1 (July 2019), pp. 353–392. ISSN: 0949-877X. DOI: 10.1007/s00778-019-00556-x.

Annotations: **05/21/26**: This is the paper we always start from for community search, with a pretty comprehensive description of algorithms and problems, and definitions. Good as a reference.

Abstract: With the rapid development of information technologies, various big graphs are prevalent in many real applications (e.g., social media and knowledge bases). An important component of these graphs is the network community. Essentially, a community is a group of vertices which are densely connected internally. Community retrieval can be used in many real applications, such as event organization, friend recommendation, and so on. Consequently, how to efficiently find high-quality communities from big graphs is an important research topic in the era of big data. Recently, a large group of research works, called community search, have been proposed. They aim to provide effciient solutions for searching high-quality communities from large networks in real time. Nevertheless, these works focus on different types of graphs and formulate communities in different manners, and thus, it is desirable to have a comprehensive review of these works. In this survey, we conduct a thorough review of existing community search works. Moreover, we analyze and compare the quality of communities under their models, and the performance of different solutions. Furthermore, we point out new research directions. This survey does not only help researchers to have better understanding of existing community search solutions, but also provides practitioners a better judgment on choosing the proper solutions.

Gou et al.: A Comprehensive Survey and Experimental Study of Learning-Based Community Search — gou2025comprehensive

Xiaoxuan Gou et al. "A Comprehensive Survey and Experimental Study of Learning-Based Community Search". In: *Proceedings of the VLDB Endowment* 18.9 (May 2025), pp. 2941–2954. ISSN: 2150-8097. DOI: 10.14778/3746405.3746419.

Abstract: Given a graph G and a query node q, the goal of community search (CS) is to find a structurally cohesive subgraph from G that contains q. Significant progress has been made in community search using deep learning in recent years. To the best of our knowledge, no existing work has provided a comprehensive review of learning-based community search methods. Additionally, we find that: (1) Existing methods offer diverse definitions or descriptions of communities, which require systematic summarization. (2) The methods rely on distinct metrics for limited community assessment. (3) Overhead evaluations of the methods vary and exhibit certain biases. Therefore, a comprehensive survey and experimental study are essential to achieve four key objectives: designing a unified pipeline, clarifying community definitions, enriching community evaluation, and establishing overhead assessment. In this paper, we first propose a unified pipeline for these methods, highlighting techniques. We categorize community definitions and analyze the relationships between identified communities. Beyond that, we proposed several community metrics to evaluate the communities comprehensively. Moreover, we introduce a more detailed overhead evaluation approach that considers resource consumption during both the training and search phases. Finally, we employ the proposed community evaluation metrics and overhead assessment framework to evaluate and analyze the methods, examine correlations among metrics, and explore the effects of several commonly used techniques.

Gou et al.: Effective and Efficient Community Search with Graph Embeddings **gou2023effective**

Xiaoxuan Gou et al. "Effective and Efficient Community Search with Graph Embeddings". In: *ECAI 2023*. IOS Press, Sept. 2023. ISBN: 9781643684376. DOI: `10.3233/faia230358`.

Annotations: **05/21/26**: k-core based community search (query sets) via graph convolutional network; requires training; I hate machine learning papers. Sometimes okay, but also cases there get $<$ 50% F1 score with true solution. maximum dataset network size is 60,000.

Abstract: . Given a graph G and a query node q , community search (CS) seeks a cohesive subgraph from G that contains q . CS has gained much research interests recently. In the database research community, researchers aim to find the most cohesive subgraph satisfying a specific community model (e.g., k -core or k -truss) via graph traversal. These works obtain good precision, however suffering from the low efficiency issue. In the AI research community, a new thought of using the deep learning model to support CS without relying on graph traversal emerges. Supervised end-to-end models using GCN are presented, which perform efficiently, but leave a large room for precision improvement. None of them can achieve a good balance between the efficiency and effectiveness. This motivates our solution: First, we present an offline community-injected graph embedding method to preserve the community's cohesiveness features into the learned node representations. Second, we resort to a proximity graph (PG) built from node representations, to quickly return the community online. Moreover, we develop a self-augmented method based on KL divergence to further optimize node representations. Extensive experiments on seven real-world graphs show our solution's superiority on effectiveness (at least 39.3% improvement) and efficiency (one to two orders of magnitude faster).

Huang et al.: Community Search over Big Graphs: Models, Algorithms, and Opportunities **huang2017community**

Xin Huang, Laks V.S. Lakshmanan, and Jianliang Xu. "Community Search over Big Graphs: Models, Algorithms, and Opportunities". In: *2017 IEEE 33rd International Conference on Data Engineering (ICDE)*. IEEE, Apr. 2017, pp. 1451–1454. DOI: `10.1109/icde.2017.211`.

Annotations: **05/21/26**: Short list of applications and problems in community search. Treat this similar to Fang et al. [5].

Jiang et al.: GNN-Based Persistent K-core Community Search in Temporal Graphs jiang2024gnnbased

Zongli Jiang et al. "GNN-Based Persistent K-core Community Search in Temporal Graphs". In: *2024 IEEE International Conference on Big Data (BigData)*. IEEE, Dec. 2024, pp. 569–578. DOI: 10.1109/bigdata62323.2024.10825281.

Abstract: The goal of community search is to provide effective solutions for real-time, high-quality community searches within large networks. In many practical applications, such as event organization and friend recommendations, discovering various community structures within a network is crucial for users. However, existing community search algorithms rarely address issues within temporal graphs, and those that do often have two main limitations: (1) traditional community search methods become inefficient and experience significant increases in computation time when scaled to large graphs; (2) while GNN-based community search methods for temporal graphs offer generalizability, they often focus solely on community connectivity and lack cohesiveness. Therefore, we propose a new model PK-GCN, based on Graph Neural Networks (GNNs), to identify persistent k-core communities in temporal networks. This model can handle dynamic changes in temporal graphs and identify communities that persist over time. Compared to existing community search methods, our model not only finds communities with tighter structures but also allows for dynamic queries based on user input without needing retraining. Specifically, our model constructs features by integrating k-core information from core decomposition, graph features, and query features, resulting in more expressive node representations. Additionally, we designed a flexible dynamic query mechanism that allows users to input time information to query communities. Experiments on multiple datasets demonstrate that our model outperforms other GNN-based community search algorithms in F1-score.

Li et al.: Efficient progressive minimum k-core search li2020efficient

Conggai Li et al. "Efficient progressive minimum k-core search". In: *Proceedings of the VLDB Endowment* (2020).

Annotations: **05/21/26**: Instead of addressing the theoretically solved *maximum* k-core search, this paper focuses on approximating (as is NP-Hard) the smallest community that has the k-core property (for k an argument to the method). Previous works uses an expansion heuristic to add nodes. This paper describes a lot of theoretical structural results (that I have not read in detail), and describes an algorithm with an approximation ratio c (user-specified), and also for query sets. It does a lot better than previous methods, and scales to graphs with billions of edges in minutes (k=10) (so this is a different problem than the one in the survey Fang et al. [5].

Abstract: As one of the most representative cohesive subgraph models, k-core model has recently received significant attention in the literature. In this paper, we investigate the problem of the minimum k-core search: given a graph G, an integer k and a set of query vertices $Q = \{q\}$, we aim to find the smallest k-core subgraph containing every query vertex $q \in Q$. It has been shown that this problem is NP-hard with a huge search space, and it is very challenging to find the optimal solution. There are several heuristic algorithms for this problem, but they rely on simple scoring functions and there is no guarantee as to the size of the resulting subgraph, compared with the optimal solution. Our empirical study also indicates that the size of their resulting subgraphs may be large in practice. In this paper, we develop an effective and efficient progressive algorithm, namely PSA, to provide a good trade-of fbetween the quality of the result and the search time. Novel lower and upper bound techniques for the minimum k-core search are designed. Our extensive experiments on 12 real-life graphs demonstrate the effectiveness and efficiency of the new techniques.

Li et al.: Influential community search in large networks li2015influential

Rong-Hua Li et al. "Influential community search in large networks". In: *Proceedings of the VLDB Endowment* 8.5 (Jan. 2015), pp. 509–520. ISSN: 2150-8097. DOI: 10.14778/2735479.2735484.

Annotations: **05/21/26**: For graphs with node weights (typically centrality), this paper describes k-influence as the (not necessarily maximal) k-core subgraph that maximizes the minimum weight, and shows that greedy peeling is optimal. They then also describe ShellStruct (under a different name).

Abstract: Community search is a problem of finding densely connected subgraphs that satisfy the query conditions in a network, which has attracted much attention in recent years. However, all the previous studies on community search do not consider the influence of a community. In this paper, we introduce a novel community model called k-influential community based on the concept of k-core, which can capture the influence of a community. Based on the new community model, we propose a linear-time online search algorithm to find the top-r k-influential communities in a network. To further speed up the influential community search algorithm, we devise a linear-space index structure which supports efficient search of the top-r k-influential communities in optimal time. We also propose an efficient algorithm to maintain the index when the network is frequently updated. We conduct extensive experiments on 7 real-world large networks, and the results demonstrate the efficiency and effectiveness of the proposed methods.

Lu et al.: On Time-optimal (k, p)-core Community Search in Dynamic Graphs lu2022timeoptimal

Zhao Lu et al. "On Time-optimal (k, p)-core Community Search in Dynamic Graphs". In: *2022 IEEE 38th International Conference on Data Engineering (ICDE)*. IEEE, May 2022, pp. 1396–1407. DOI: `10.1109/icde53745.2022.00109`.

Annotations: **05/21/26**: Describes a new notion of (k, p)-Core, where it is a k-core such that p fraction of its neighbors in the subgraph. the maximal subgraph (solution to their proposed problem) may not be connected. They propose an index (using anchored union-find, similar to Barbieri et al. [2] that supports dynamic updates. time-optimal refers to queries, not the index construction.

Abstract: Community search aims to find cohesive subgraphs containing certain vertices, attracting increasing interest recently. However, existing cohesive models such as k-core mainly focus on the dense connections inside the community, and neglect the interactions with the vertices outside. In this paper, we study the (k,p) -core community search (KPCS) problem in dynamic graphs, i.e., find the maximal connected subgraph containing a query vertex where each vertex has at least k neighbors and at least p fraction of its neighbors in the subgraph. Such fraction and connectivity constraints bring non-trivial challenges to the online community search in dynamic graphs. Thus, we design a space-efficient $O(m)$ where m is the edge number) index KPForest which can support time-optimal (k,p) -core community search. We also propose novel construction and maintenance algorithms to record and update the (k,p) value and the connectivity information for dynamic graphs correctly and efficiently. Extensive experimental studies on ten real-world datasets show that our index can support community search with two orders of magnitude speedup at a small cost of construction and maintenance compared with the baseline algorithms.

O'Brien et al.: Locally Estimating Core Numbers obrien2014locally

Michael P. O'Brien and Blair D. Sullivan. "Locally Estimating Core Numbers". In: *2014 IEEE International Conference on Data Mining*. IEEE, Dec. 2014, pp. 460–469. DOI: `10.1109/icdm.2014.136`.

Annotations: **05/18/26**: This paper develops very good theory for neighborhood-based estimation of the core number. Mathematically precise with proofs, but easy to read and good examples/figures. They show empirically that a constant number of steps is a very good approximation, and theoretically it converges to the true coreness.

Abstract: Graphs are a powerful way to model interactions and relationships in data from a wide variety of application domains. In this setting, entities represented by vertices at the 'center' of the graph are often more important than those associated with vertices on the 'fringes'. For example, central nodes tend to be more critical in the spread of information or disease and play an important role in clustering/community formation. Identifying such 'core' vertices has recently received additional attention in the context of network experiments, which analyze the response when a random subset of vertices are exposed to a treatment (e.g. Inoculation, free product samples, etc). Specifically, the likelihood of having many central vertices in any exposure subset can have a significant impact on the experiment. We focus on using k-cores and core numbers to measure the extent to which a vertex is central in a graph. Existing algorithms for computing the core number of a vertex require the entire graph as input, an unrealistic scenario in many real world applications. Moreover, in the context of network experiments, the sub graph induced by the treated vertices is only known in a probabilistic sense. We introduce a new method for estimating the core number based only on the properties of the graph within a region of radius δ around the vertex, and prove an asymptotic error bound of our estimator on random graphs. Further, we empirically validate the accuracy of our estimator for small values of δ on a representative corpus of real data sets. Finally, we evaluate the impact of improved local estimation on an open problem in network experimentation posed by Ugander et al.

Sarıyüce et al.: Incremental k-core decomposition: algorithms and evaluation saryce2016incremental

Ahmet Erdem Sarıyüce et al. "Incremental k-core decomposition: algorithms and evaluation". In: *The VLDB Journal* 25.3 (Feb. 2016), pp. 425–447. ISSN: 0949-877X. DOI: `10.1007/s00778-016-0423-8`.

Annotations: **05/18/26**: This paper develops local traversal style algorithms that exactly update the coreness after an edge insertion/deletion in linear time. It does a lot of good pruning, but makes no theoretical progress, as in the worst case, all core numbers can change (e.g. adding an edge to turn path to cycle).

Abstract: A k-core of a graph is a maximal connected subgraph in which every vertex is connected to at least k vertices in the subgraph. k-core decomposition is often used in large-scale network analysis, such as community detection, protein function prediction, visualization, and solving NP-hard problems on real networks efficiently, like maximal clique finding. In many real-world applications, networks change over time. As a result, it is essential to develop efficient incremental algorithms for dynamic graph data. In this paper, we propose a suite of incremental k-core decomposition algorithms for dynamic graph data. These algorithms locate a small subgraph that is guaranteed to contain the list of vertices whose maximum k-core values have changed and efficiently process this subgraph to update the k-core decomposition. We present incremental algorithms for both insertion and deletion operations, and propose auxiliary vertex state maintenance techniques that can further accelerate these operations. Our results show a significant reduction in runtime compared to non-incremental alternatives. We illustrate the efficiency of our algorithms on different types of real and synthetic graphs, at varying scales. For a graph of 16 million vertices, we observe relative throughputs reaching a million times, relative to the non-incremental algorithms.

Sozio et al.: The community-search problem and how to plan a successful cocktail party sozio2010communitysearch

Mauro Sozio and Aristides Gionis. "The community-search problem and how to plan a successful cocktail party". In: *Proceedings of the 16th ACM SIGKDD international conference on Knowledge discovery and data mining*. KDD '10. ACM, July 2010, pp. 939–948. DOI: `10.1145/1835804.1835923`.

Annotations: **05/21/26**: This is the original paper for community search. The problem is to find the maximium score, connected (not necessarily the largest size community). They also often add a distance constraint. Describes the Local and Global search problem, and an exact greedy algorithm that optimizes a small family of objectives.

Abstract: A lot of research in graph mining has been devoted in the dis-covery of communities. Most of the work has focused in thescenario where communities need to be discovered with onlyreference to the input graph. However, for many interest-ing applications one is interested in finding the communityformed by a given set of nodes. In this paper we study aquery-dependent variant of the community-detection prob-lem, which we call the community-search problem: given agraph G, and a set of query nodes in the graph, we seek tofind a subgraph of G that contains the query nodes and itis densely connected.We motivate a measure of

density based on minimum de-gree and distance constraints, and we develop an optimumgreedy algorithm for this measure. We proceed by charac-terizing a class of monotone constraints and we generalizeour algorithm to compute optimum solutions satisfying anyset of monotone constraints. Finally we modify the greedyalgorithm and we present two heuristic algorithms that findcommunities of size no greater than a specified upper bound.Our experimental evaluation on real datasets demonstratesthe efficiency of the proposed algorithms and the quality ofthe solutions we obtain.

Sun et al.: Index-Based Intimate-Core Community Search in Large Weighted Graphs sun2022indexbased

Longxu Sun et al. "Index-Based Intimate-Core Community Search in Large Weighted Graphs". In: *IEEE Transactions on Knowledge and Data Engineering* 34.9 (Sept. 2022), pp. 4313–4327. ISSN: 2326-3865. DOI: `10.1109/tkde.2020.3040762`.

Abstract: Community search that finds query-dependent communities has been studied on various kinds of graphs. As one instance of community search, intimate-core group (community) search over a weighted graph is to find a connected ¡inline-formula¿¡tex-math notation="LaTeX"¿k¡/tex-math¿¡alternatives¿¡mml:math¿¡mml:mi¿k¡/mml:n graphic xlink:href="sun-ieq1-3040762.gif"/¿¡/alternatives¿¡/inline-formula¿-core containing all query nodes with the smallest group weight. However, existing state-of-the-art methods start from the maximal ¡inline-formula¿¡tex-math notation="LaTeX"¿k¡/tex-math¿¡alternatives¿¡mml:math¿¡mml:mi¿k¡/mml:mi¿¡/mml:math¿¡inline-graphic xlink:href="sun-ieq2-3040762.gif"/¿¡/alternatives¿¡/inline-formula¿-core to refine an answer, which is practically inefficient for large networks. In this paper, we develop an efficient framework, called ¡underline¿l¡/underline¿ocal ¡underline¿e¡/underline¿xploration ¡underline¿k¡/underline¿-core ¡underline¿s¡/underline¿earch (LEKS), to find intimate-core groups in graphs. We propose a small-weighted spanning tree to connect query nodes, and then expand the tree level by level to a connected ¡inline-formula¿¡tex-math notation="LaTeX"¿k¡/tex-math¿¡alternatives¿¡mml:math¿¡mml:mi¿k¡/mml:mi¿¡/mml:math¿¡inline-graphic xlink:href="sun-ieq3-3040762.gif"/¿¡/altern formula¿-core, which is finally refined as an intimate-core group. In addition, to support the intimate group search over large weighted graphs, we develop a weighted-core index (WC-index) and two new WC-index-based algorithms for expansion and refinement phases in LEKS. Specifically, we propose a WC-index-based expansion to efficiently find a candidate graph of intimate-core group, leveraging on a two-level expansion of ¡inline-formula¿¡tex-math notation="LaTeX"¿k¡/tex-math¿¡alternatives¿¡mml:math¿¡mml:mi¿k¡/mml:mi¿¡/mml:math¿¡inline graphic xlink:href="sun-ieq4-3040762.gif"/¿¡/alternatives¿¡/inline-formula¿-breadth and 1-depth. We propose two graph removal strategies: coarse-grained refinement is designed for large graphs to delete a batch of nodes in a few iterations; fine-grained refinement is designed for small graphs to remove nodes carefully and achieve high-quality answers. Extensive experiments on real-life networks with ground-truth communities validate the effectiveness and efficiency of our proposed methods.

Tang et al.: Reliable community search in dynamic networks tang2022reliable

Yifu Tang et al. "Reliable community search in dynamic networks". In: *Proceedings of the VLDB Endowment* 15.11 (July 2022), pp. 2826–2838. ISSN: 2150-8097. DOI: `10.14778/3551793.3551834`.

Abstract: Searching for local communities is an important research problem that supports advanced data analysis in various complex networks, such as social networks, collaboration networks, cellular networks, etc. The evolution of such networks over time has motivated several recent studies to identify local communities in dynamic networks. However, these studies only utilize the aggregation of disjoint structural information to measure the quality and ignore the reliability of the communities in a continuous time interval. To fill this research gap, we propose a novel (θ, k)-core reliable community (CRC) model in the weighted dynamic networks, and define the problem of most reliable community search that couples the desirable properties of connection strength, cohesive structure continuity, and the maximal member engagement. To solve this problem, we first develop a novel edge filtering based online CRC search algorithm that can effectively filter out the trivial edge information from the networks while searching for a reliable community. Further, we propose an index structure, Weighted Core Forest-Index (WCF-index), and devise an index-based dynamic

programming CRC search algorithm, that can prune a large number of insignificant intermediate results and support efficient query processing. Finally, we conduct extensive experiments systematically to demonstrate the efficiency and effectiveness of our proposed algorithms on eight real datasets under various experimental settings.

Wen et al.: I/O Efficient Core Graph Decomposition: Application to Degeneracy Ordering wen2019io

Dong Wen et al. "I/O Efficient Core Graph Decomposition: Application to Degeneracy Ordering". In: *IEEE Transactions on Knowledge and Data Engineering* 31.1 (Jan. 2019), pp. 75–90. ISSN: 2326-3865. DOI: 10.1109/tkde.2018.2833070.

Annotations: **05/19/26**: Propagates an upper-bound for the coreness, a function of just a node's neighbors. Hence, with graph partitioning, we can scale k-core estimation to just use $O(n)$ space. The notation and writing is a bit clunky, but is overall pretty good work.

Abstract: Core decomposition is a fundamental graph problem with a large number of applications. Most existing approaches for core decomposition assume that the graph is kept in memory of a machine. Nevertheless, many real-world graphs are too big to reside in memory. In this paper, we study I/O efficient core decomposition following a semi-external model, which only allows node information to be loaded in memory. We propose a semi-external algorithm and an optimized algorithm for I/O efficient core decomposition. To handle dynamic graph updates, we firstly show that our algorithm can be naturally extended to handle edge deletion. Then, we propose an I/O efficient core maintenance algorithm to handle edge insertion, and an improved algorithm to further reduce I/O and CPU cost. In addition, based on our core decomposition algorithms, we further propose an I/O efficient semi-external algorithm for degeneracy ordering, which is an important graph problem that is highly related to core decomposition. We also consider how to maintain the degeneracy order. We conduct extensive experiments on 12 real large graphs. Our optimal core decomposition algorithm significantly outperforms the existing I/O efficient algorithm in terms of both processing time and memory consumption. They are very scalable to handle web-scale graphs. As an example, we are the first to handle a web graph with 978.5 million nodes and 42.6 billion edges using less than 4.2 GB memory. We also show that our proposed algorithms for degeneracy order computation and maintenance can handle big graphs efficiently with small memory overhead.

Zhang et al.: Efficient community discovery with user engagement and similarity zhang2019efficient

Fan Zhang et al. "Efficient community discovery with user engagement and similarity". In: *The VLDB Journal* 28.6 (Oct. 2019), pp. 987–1012. ISSN: 0949-877X. DOI: 10.1007/s00778-019-00579-4.

Annotations: **05/21/26**: Propose a new objective, (k, r)-core, where k is the coreness and r is a threshold that all nodes within the community must have pairwise similarity at least r. NP-Hard. Lots of heuristics. Lots of good examples, too, but very long paper, although mostly well-written (although I am not interested).

Abstract: In this paper, we investigate the problem of (k,r)-core which intends to find cohesive subgraphs on social networks considering both user engagement and similarity perspectives. In particular, we adopt the popular concept of k-core to guarantee the engagement of the users (vertices) in a group (subgraph) where each vertex in a (k,r)-core connects to at least k other vertices. Meanwhile, we consider the pairwise similarity among users based on their attributes. Efficient algorithms are proposed to enumerate all maximal (k,r)-cores and find the maximum (k,r)-core, where both problems are shown to be NP-hard. Effective pruning techniques substantially reduce the search space of two algorithms. A novel (k,k')-core based (k,r)-core size upper bound enhances the performance of the maximum (k,r)-core computation. We also devise effective search orders for two algorithms with different search priorities for vertices. Besides, we study the diversified (k,r)-core search problem to find l maximal (k,r)-cores which cover the most vertices in total. These maximal

(k,r)-cores are distinctive and informationally rich. An efficient algorithm is proposed with a guaranteed approximation ratio. We design a tight upper bound to prune unpromising partial (k,r)-cores. A new search order is designed to speed up the search. Initial candidates with large size are generated to further enhance the pruning power. Comprehensive experiments on real-life data demonstrate that the maximal (k,r)-cores enable us to find interesting cohesive subgraphs, and performance of three mining algorithms is effectively improved by all the proposed techniques.

Community Detection

Fortunato et al.: Resolution limit in community detection **fortunato2007resolution**

Santo Fortunato and Marc Barthélemy. "Resolution limit in community detection". In: *Proceedings of the National Academy of Sciences* 104.1 (Jan. 2007), pp. 36–41. ISSN: 1091-6490. DOI: `10.1073/pnas.0605965104`.

Annotations: **05/18/26**: This paper addresses the $\sqrt{E}$ volume bias of modularity optimization. For two communities of volume $< \sqrt{2E}$, then if they are connected by even one edge, modularity is strictly increased after merging, and we conclude a limit of $\sqrt{2E}$ communities in the optimal modularity solution. This is a short, well-written paper with instructive examples, e.g. ring-of-cliques.

Abstract: Detecting community structure is fundamental for uncovering the links between structure and function in complex networks and for practical applications in many disciplines such as biology and sociology. A popular method now widely used relies on the optimization of a quantity called modularity, which is a quality index for a partition of a network into communities. We find that modularity optimization may fail to identify modules smaller than a scale which depends on the total size of the network and on the degree of interconnectedness of the modules, even in cases where modules are unambiguously defnied. This finding is confirmed through several examples, both in artificial and in real social, biological, and technological networks, where we show that modularity optimization indeed does not resolve a large number of modules. A check of the modules obtained through modularity optimization is thus necessary, and we provide here key elements for the assessment of the reliability of this community detection method.

Rosvall et al.: Maps of random walks on complex networks reveal community structure **rosvall2008maps**

Martin Rosvall and Carl T. Bergstrom. "Maps of random walks on complex networks reveal community structure". In: *Proceedings of the National Academy of Sciences* 105.4 (Jan. 2008), pp. 1118–1123. ISSN: 1091-6490. DOI: `10.1073/pnas.0706851105`.

Annotations: **05/18/26**: Very interesting paper with a distinctive approach to clustering. It defines what it wants to capture (information flow), substitutes a reasonable surrogate (probability flow), and very elegantly describes an optimization function (description length via multi-layer encoding) to capture it. Lots of very instructive diagrams, and overall very nicely written.

Abstract: To comprehend the multipartite organization of large-scale biological and social systems, we introduce an information theoretic approach that reveals community structure in weighted and directed networks. We use the probability flow of random walks on a network as a proxy for information flows in the real system and decompose the network into modules by compressing a description of the probability flow. The result is a map that both simplifies and highlights the regularities in the structure and their relationships. We illustrate the method by making a map of scientific communication as captured in the citation patterns of >6,000 journals. We discover a multicentric organization with fields that vary dramatically in size and degree of integration into the network of science. Along the backbone of the network—including physics, chemistry, molecular biology, and medicine—information flows bidirectionally, but the map reveals a directional pattern of citation from the applied fields to the basic sciences.

Traag et al.: Narrow scope for resolution-limit-free community detection **traag2011narrow**

V. A. Traag, P. Van Dooren, and Y. Nesterov. "Narrow scope for resolution-limit-free community detection". In: *Physical Review E* 84.1 (July 2011). ISSN: 1550-2376. DOI: `10.1103/physreve.84.016114`.

Annotations: **05/19/26**: If a resolution-limit free algorithm produces a clustering $\mathcal{C} = \{C_i\}$, then it should produce the same clustering on any induced subgraph of a union of clusters. Traag et al. describes *clustering* quality objectives, and show that *only* their proposed objective CPM, or something very similar to it, can be resolution limit free. Finally, they perform experiments showing good empirical performance. This paper seems OK; I don't like the problem of having to choose the right resolutin parameter. Good theory, and much much better than modularity. Some more examples and intuition behind CPM would be good, but is provided in the Leiden suppl.

Abstract: Detecting communities in large networks has drawn much attention over the years. While modularity remains one of the more popular methods of community detection, the so-called resolution limit remains a significant drawback. To overcome this issue, it was recently suggested that instead of comparing the network to a random null model, as is done in modularity, it should be compared to a constant factor. However, it is unclear what is meant exactly by "resolution-limit-free," that is, not suffering from the resolution limit. Furthermore, the question remains what other methods could be classified as resolution-limit-free. In this paper we suggest a rigorous definition and derive some basic properties of resolution-limit-free methods. More importantly, we are able to prove exactly which class of community detection methods are resolution-limit-free. Furthermore, we analyze which methods are not resolution-limit-free, suggesting there is only a limited scope for resolution-limit-free community detection methods. Finally, we provide such a natural formulation, and show it performs superbly.

Community Evaluation

Vu-Le et al.: EC-SBM synthetic network generator **vu-le2025ecsbm**

The-Anh Vu-Le et al. "EC-SBM synthetic network generator". In: *Applied Network Science* 10.1 (May 2025). ISSN: 2364-8228. DOI: 10.1007/s41109-025-00701-2.

Annotations: **05/19/26**: A high quality network generator should have similar statistics to empirical networks, and be able to scale to large networks. In particular, they want to recover community structure, such as clustering coefficient and connectivity of communities. To ensure connectivity, they overlay a k-edge connected spanning subnetwork with an SBM network, with final step of degree correction, finding the best parameters from a SBM+WCC clustering. They perform the best on all metrics (compare to RECCS, SBM, ...) except connectivity, with RECCS being a bit better. This paper is quite long and has a comprehensive set of experiments; I wonder what the network community profile looks like. Seems okay to use as a blackbox and not think about its mechanics.

Abstract: Generating high-quality synthetic networks with realistic community structure is vital to effectively evaluate community detection algorithms. In this study, we propose a new synthetic network generator called the Edge-Connected Stochastic Block Model (EC-SBM). The goal of EC-SBM is to take a given clustered real-world network and produce a synthetic network that resembles the clustered real-world network with respect to both network and community-specific criteria. In particular, we focus on simulating the internal edge connectivity of the clusters in the reference clustered network. Our performance study on large real-world networks shows that EC-SBM is generally more accurate with respect to network and community criteria than currently used approaches for this problem. Furthermore, we demonstrate that EC-SBM can complete analyses on several real-world networks with millions of nodes.

Leskovec et al.: Empirical comparison of algorithms for network community detection **leskovec2010empirical**

Jure Leskovec, Kevin J. Lang, and Michael Mahoney. "Empirical comparison of algorithms for network community detection". In: *Proceedings of the 19th international conference on World wide web*. WWW '10. ACM, Apr. 2010, pp. 631–640. DOI: 10.1145/1772690.1772755.

Annotations: **05/21/26**: This paper is closely related to the evaluation in Yang and Leskovec [25]. Using the same objective functions, they evaluate various clustering algorithms on. This paper also considers size resolved cluster quality, i.e. the NCP, typically a V shape on log-log scales. This paper gives another set of examples for *discriminative* qualities (bias towards certain types of clusters), and inductive bias of algorithms (e.g. separability vs compactness, even for same objective). This paper is okay; its a good step towards bridging theory and practice, but applied to flow/spectral algorithms which I don't know much about.

Abstract: Detecting clusters or communities in large real-world graphs such as large social or information networks is a problem of considerable interest. In practice, one typically chooses an objective function that captures the intuition of a network cluster as set of nodes with better internal connectivity than external connectivity, and then one applies approximation algorithms or heuristics to extract sets of nodes that are related to the objective function and that "look like" good communities for the application of interest. In this paper, we explore a range of network community detection methods in order to compare them and to understand their relative performance and the systematic biases in the clusters they identify. We evaluate several common objective functions that are used to formalize the notion of a network community, and we examine several different classes of approximation algorithms that aim to optimize such objective functions. In addition, rather than simply fixing an objective and asking for an approximation to the best cluster of any size, we consider a size-resolved version of the optimization problem. Considering community quality as a function of its size provides a much finer lens with which to examine community detection algorithms, since objective functions and approximation algorithms often have non-obvious size-dependent behavior.

Yang et al.: Defining and evaluating network communities based on ground-truth
yang2012defining

Jaewon Yang and Jure Leskovec. "Defining and evaluating network communities based on ground-truth". In: *Proceedings of the ACM SIGKDD Workshop on Mining Data Semantics*. KDD '12. ACM, Aug. 2012, pp. 1–8. DOI: 10.1145/2350190.2350193.

Annotations: **05/21/26**: This paper allows for overlapping clusters, as they are scored individually, and split by components. Distinction between goodness and scoring is still a bit confusing – whats wrong with using goodness as a scoring criterion? Also the cohesiveness formula is a bit strange, but similar to NCP, and perhaps densest subgraph. The clustering coefficient is defined as $g(S) = \frac{1}{|S|}\sum_{v \in S} \frac{2|E(N(v))|}{|N(v)|\cdot(|N(v)|-1)}$, it and density are similar to tpr. Cohesiveness is defined as maximum subgraph conductance, it and separability are similar to conductance. Desirable traits are stable under small perturbations, but sharp increase with large perturbation — evaluation familiar to Fisher information (more sensitive is more information). 6 distinct sources of networks (225/230 of them are from same platform). *Thought experiment: suppose clustering is inaccurate. What does this paper tell us?* We don't know the node coverage. **05/18/26**: There are three barriers for evaluating accuracy of clustering algorithms on empirical: 1. lack of datasets, 2. lack of reliable ground-truth, and 3. a common definition for what a community is. The authors propose 4 axioms (community goodness) that a good community is, namely it must be separable, dense, cohesive, and have high clustering coefficient. They evaluate 13 scoring functions, finding that modularity tends to be negatively correlated with a good community and suggest joint evaluation using conductance and TPR for robust evaluation. Finally, using these functions, they procure a set of datasets with (somewhat) reliable ground-truth. This paper is OK; takeaway is their procedure of evaluating scoring methods, and the ridiculing of modularity, but the ground-truth they create from them is questionable.

Abstract: Nodes in real-world networks, such as social, information or technological networks, organize into communities where edges appear with high concentration among the members of the community. Identifying communities in networks has proven to be a challenging task mainly due to a plethora of definitions of a community, intractability of algorithms, issues with evaluation and the lack of a reliable gold-standard ground-truth. We study a set of 230 large social, collaboration and information networks where nodes explicitly define group memberships. We use these groups to define the notion of ground-truth communities. We then propose a methodology which allows us to compare and quantitatively evaluate different definitions of network communities on a large scale. We choose 13 commonly used definitions of network communities and examine their quality, sensitivity and robustness. We show that the 13 definitions naturally group into four classes. We find that two of these definitions, Conductance and Triad-participation-ratio, consistently give the best performance in identifying ground-truth communities.

Marked for Re-Reading

Gou et al.: A Comprehensive Survey and Experimental Study of Learning-Based Community Search gou2025comprehensive

Xiaoxuan Gou et al. "A Comprehensive Survey and Experimental Study of Learning-Based Community Search". In: *Proceedings of the VLDB Endowment* 18.9 (May 2025), pp. 2941–2954. ISSN: 2150-8097. DOI: 10.14778/3746405.3746419.

Abstract: Given a graph G and a query node q, the goal of community search (CS) is to find a structurally cohesive subgraph from G that contains q. Significant progress has been made in community search using deep learning in recent years. To the best of our knowledge, no existing work has provided a comprehensive review of learning-based community search methods. Additionally, we find that: (1) Existing methods offer diverse definitions or descriptions of communities, which require systematic summarization. (2) The methods rely on distinct metrics for limited community assessment. (3) Overhead evaluations of the methods vary and exhibit certain biases. Therefore, a comprehensive survey and experimental study are essential to achieve four key objectives: designing a unified pipeline, clarifying community definitions, enriching community evaluation, and establishing overhead assessment. In this paper, we first propose a unified pipeline for these methods, highlighting techniques. We categorize community definitions and analyze the relationships between identified communities. Beyond that, we proposed several community metrics to evaluate the communities comprehensively. Moreover, we introduce a more detailed overhead evaluation approach that considers resource consumption during both the training and search phases. Finally, we employ the proposed community evaluation metrics and overhead assessment framework to evaluate and analyze the methods, examine correlations among metrics, and explore the effects of several commonly used techniques.

Gou et al.: Effective and Efficient Community Search with Graph Embeddings gou2023effective

Xiaoxuan Gou et al. "Effective and Efficient Community Search with Graph Embeddings". In: *ECAI 2023*. IOS Press, Sept. 2023. ISBN: 9781643684376. DOI: 10.3233/faia230358.

Annotations: **05/21/26**: k-core based community search (query sets) via graph convolutional network; requires training; I hate machine learning papers. Sometimes okay, but also cases there get $<$ 50% F1 score with true solution. maximum dataset network size is 60,000.

Abstract: . Given a graph G and a query node q , community search (CS) seeks a cohesive subgraph from G that contains q . CS has gained much research interests recently. In the database research community, researchers aim to find the most cohesive subgraph satisfying a specific community model (e.g., k -core or k -truss) via graph traversal. These works obtain good precision, however suffering from the low efficiency issue. In the AI research community, a new thought of using the deep learning model to support CS without relying on graph traversal emerges. Supervised end-to-end models using GCN are presented, which perform efficiently, but leave a large room for precision improvement. None of them can achieve a good balance between the efficiency and effectiveness. This motivates our solution: First, we present an offline community-injected graph embedding method to preserve the community's cohesiveness features into the learned node representations. Second, we resort to a proximity graph (PG) built from node representations, to quickly return the community online. Moreover, we develop a self-augmented method based on KL divergence to further optimize node representations. Extensive experiments on seven real-world graphs show our solution's superiority on effectiveness (at least 39.3% improvement) and efficiency (one to two orders of magnitude faster).

Sozio et al.: The community-search problem and how to plan a successful cocktail party sozio2010communitysearch

Mauro Sozio and Aristides Gionis. "The community-search problem and how to plan a successful cocktail party". In: *Proceedings of the 16th ACM SIGKDD international conference on Knowledge discovery and data mining*. KDD '10. ACM, July 2010, pp. 939–948. DOI: 10.1145/1835804.1835923.

Annotations: **05/21/26**: This is the original paper for community search. The problem is to find the maximium score, connected (not necessarily the largest size community). They also often add a distance constraint. Describes the Local and Global search problem, and an exact greedy algorithm that optimizes a small family of objectives.

Abstract: A lot of research in graph mining has been devoted in the dis-covery of communities. Most of the work has focused in thescenario where communities need to be discovered with onlyreference to the input graph. However, for many interest-ing applications one is interested in finding the communityformed by a given set of nodes. In this paper we study aquery-dependent variant of the community-detection prob-lem, which we call the community-search problem: given agraph G, and a set of query nodes in the graph, we seek tofind a subgraph of G that contains the query nodes and itis densely connected.We motivate a measure of density based on minimum de-gree and distance constraints, and we develop an optimumgreedy algorithm for this measure. We proceed by charac-terizing a class of monotone constraints and we generalizeour algorithm to compute optimum solutions satisfying anyset of monotone constraints. Finally we modify the greedyalgorithm and we present two heuristic algorithms that findcommunities of size no greater than a specified upper bound.Our experimental evaluation on real datasets demonstratesthe efficiency of the proposed algorithms and the quality ofthe solutions we obtain.

Tang et al.: Reliable community search in dynamic networks **tang2022reliable**

Yifu Tang et al. "Reliable community search in dynamic networks". In: *Proceedings of the VLDB Endowment* 15.11 (July 2022), pp. 2826–2838. ISSN: 2150-8097. DOI: 10.14778/3551793.3551834.

Abstract: Searching for local communities is an important research problem that supports advanced data analysis in various complex networks, such as social networks, collaboration networks, cellular networks, etc. The evolution of such networks over time has motivated several recent studies to identify local communities in dynamic networks. However, these studies only utilize the aggregation of disjoint structural information to measure the quality and ignore the reliability of the communities in a continuous time interval. To fill this research gap, we propose a novel (θ, k)-core reliable community (CRC) model in the weighted dynamic networks, and define the problem of most reliable community search that couples the desirable properties of connection strength, cohesive structure continuity, and the maximal member engagement. To solve this problem, we first develop a novel edge filtering based online CRC search algorithm that can effectively filter out the trivial edge information from the networks while searching for a reliable community. Further, we propose an index structure, Weighted Core Forest-Index (WCF-index), and devise an index-based dynamic programming CRC search algorithm, that can prune a large number of insignificant intermediate results and support efficient query processing. Finally, we conduct extensive experiments systematically to demonstrate the efficiency and effectiveness of our proposed algorithms on eight real datasets under various experimental settings.

Watts et al.: Collective dynamics of 'small-world' networks **watts1998collective**

Duncan J. Watts and Steven H. Strogatz. "Collective dynamics of 'small-world' networks". In: *Nature* 393.6684 (June 1998), pp. 440–442. ISSN: 1476-4687. DOI: 10.1038/30918.

Annotations: **05/21/26**: This short article defines the clustering coefficient (average over vertices of edge density in neighborhoods) and the characteristic time (average of APSP matrix). This invents the notion of small-world networks and proposes a (parameterized) random process that can describe the behavior of these stats. They use this model to investigate functional traits, such as disease spread. Apparently this model is very different from previous models – second look would be good.

Abstract: Networks of coupled dynamical systems have been used to model biological oscillators, Josephson junction arrays, excitable media, neural networks, spatial games, genetic control networks and many other

self-organizing systems. Ordinarily, the connection topology is assumed to be either completely regular or completely random. But many biological, technological and social networks lie somewhere between these two extremes. Here we explore simple models of networks that can be tuned through this middle ground: regular networks 'rewired' to introduce increasing amounts of disorder. We find that these systems can be highly clustered, like regular lattices, yet have small characteristic path lengths, like random graphs. We call them 'small-world' networks, by analogy with the small-world phenomenon (popularly known as six degrees of separation). The neural network of the worm Caenorhabditis elegans, the power grid of the western United States, and the collaboration graph of film actors are shown to be small-world networks. Models of dynamical systems with small-world coupling display enhanced signal-propagation speed, computational power, and synchronizability. In particular, infectious diseases spread more easily in small-world networks than in regular lattices.

Miscellaneous

Watts et al.: Collective dynamics of 'small-world' networks — watts1998collective

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